清华大学本科生考试试题专用纸

考试课程：概率论　　考试时间：2020 年 11 月 11 日 13:30-16:05

姓名:_____　学号:_____　院系名称:_____　成绩:_____

请注意：本试卷包括十个大题，满分 100 分.

一. (18 分) 判断题：在概率空间 $(\Omega, \mathscr{F}, \mathbb{P})$ 中，

(1) (2 分) 已知 A 是一个事件，若对任意 $w \in A$，皆有 $P(w) = 0$, 则 $P(A) = 0$;

(2) (2 分) 设随机变量 X 服从泊松分布，$X \sim \mathcal{P}(\lambda)$，则有 $\mathbb{P}(X = k+1 | X > k) = \mathbb{P}(X = 1)$.

(3) (2 分) 如果 (U, V) 是连续型随机向量, 则 $P(U = V) = 0$.

(4) (2 分) 假设有随机变量 $X \sim \mathcal{N}(0, 1)$, 和随机变量 $Y = X$. 则 $P(X = Y) = 0$.

(5) (2 分) 已知 A,B, C 都是事件，若有 $P(A \cap B \cap C) = P(A)P(B)P(C)$, 则 A, B, C 相互独立.

(6) (2 分) 已知 A,B 都是事件，若有 $A \cap B = \emptyset$，则 A, B 独立;

(7) (2 分) 若 X, Y, Z 是该概率空间上的相互独立的三个离散型随机变量，则 $X + Y$ 与 Z 相互独立;

(8) (2 分) 已知 A,B 都是事件，若有 $P(B) > 0$, 则 $P(A|B) \geq P(A)$.

(9) (2 分) 已知事件 A 发生，则 $P(A) = 1$.

二. (8 分)

设随机变量 X 有如下分布函数

$$F(x) = \begin{cases} 0, & \text{if} \quad x < -1, \\ \frac{1}{2}x + \frac{1}{2}, & \text{if} \quad -1 \leq x < 0, \\ \frac{3}{4} + \frac{1}{4}x, & \text{if} \quad 0 \leq x < 1, \\ 1, & \text{if} \quad x \geq 1. \end{cases}$$

请 (1)(2 分) 画出该分布函数，并求: (2)(2 分) $P(X = -1)$, (3)(2 分) $P(X = 0)$, (4)(2 分) $P(X > \frac{1}{2})$.

三. (6 分) 设顾客在某银行的窗口等待服务的时间 X(以 min 计) 服从指数分布 $\mathcal{E}(\frac{1}{5})$. 某顾客一共去了 4 次该银行，每次等待的时间是相互独立的. 求至少有 1 次的等待时间小于 10 分钟 (min) 的概率.(可以用不计算器，化简到最后一步即可)

四. (10 分)

(1)(5 分) 设 $X \sim \mathcal{E}(\lambda)$, 求 $Y = 2X + 1$ 的概率密度函数.

(2)(5 分) 若 $X \sim \mathcal{U}(-1, 1)$, 求 $Y = |3X|$ 的概率密度函数.

五. (15 分) 设 (X,Y) 有联合概率密度函数 (Joint PDF)

$$f(x,y)=\begin{cases}6x^2e^{-2y}, & \text{if} \quad 0\le x\le 1, y>0,\\ 0, & \text{otherwise.}\end{cases}$$

(1)(5 分) 请问 X,Y 是否相互独立? 若独立，请证明你的结论；若不独立，请说明理由.

(2)(5 分) 求 X 的边缘分布函数 (Marginal CDF)$F_X(x)$.

(3)(5 分) 请给出 Y 服从的分布的名字和相应参数.

六. (14 分) 某课程共有 N 位同学，其中分别有大一、大二、大三、大四各 N_1, N_2,N_3,N_4 位同学. $N_i>0, i=1,...,4$, 且 $\sum_{i=1}^4 N_i=N$. 某点名系统在每次点名时，都将这 N 位同学的名字随机打乱，随机抽取. 某老师计划一堂课一共点 n 次名，记 X_i 为大 i 的同学出现的次数, $i=1,...,4$.

(1)(4 分) 请写出 X_1,...,X_4 的联合分布列 $P_{X_1,...,X_4}(k_1,...,k_4)$.

(2)(4 分) 请写出 (X_1, X_2) 的边缘分布列.

(3)(3 分) 请问 X_1,X_2 是否相互独立? 若独立，请证明；若不独立，请说明理由.

(4)(3 分) 现点名系统进行改革，每次被点到的同学不再进入点名库，即不会重复点同一个同学. 当一共点了 r 位同学时，记 Y 为其中总共点到的大二同学的人数. 其中 $r\le N$. 请写出 Y 的分布列, 及推导过程.

七. (9 分) 在半圆 $\{(x,y)|x^2+y^2\le r, y\ge 0\}$ 内按均匀分布随机地取一个点 (X,Y) (这里 $r>0$ 是固定的正数).

(1)(3 分) 请求出 (X,Y) 的联合概率密度函数 (Joint PDF).

(2)(3 分) 请求出给定 $Y=y$ 时，X 的条件概率密度函数 $f_{X|Y}(x|y)$.

(3)(3 分) 请问 X,Y 是否相互独立? 若独立，请证明你的结论；若不独立，请说明理由.

八. (10 分) 设某试验的样本空间为 $\Omega=[0,1]$. 考察事件 $A_n=(\frac{1}{3^n},\frac{1}{3^{n-1}}]$, $n=1,2,....$

(1)(4 分) 请写出由 $\{A_2\}$ 生成的事件域 $\mathscr{F}$.

(2)(3 分) 定义 $\mathbb{P}$ 为区间长度. 求 $P(\lim\limits_{n\to\infty}\cup_{i=1}^n A_i)$.

(3)(3 分) 求 $\limsup\limits_{n\to\infty} A_n$.

九. (5 分) 设对于某疾病，人群中患有该疾病的概率为 0.1. 现在有东、西两家医院都推出了准确率为 0.9 的检验方法: 如果一个被检的人有此病，其检查结果为阳性的概率为 0.9；如果一个被检的人没有此病，其检查结果为阴性的概率为 0.9. 同时假设，无论被检者是否有病，两家医院的检查结果都是相互独立的.

(1)(3 分) 请求出“两家医院检查结果都是阳性的情况下，被检者确实有此病”的概率.

(2)(2 分) Bob 先去了西边医院，检查结果为阴性，又去了东边医院检查结果为阳性. 他综合两个检查结果判断自己有此病的概率 p 时，后悔没有先去东边再去西边. 请你帮助他分析，是否去不同医院的顺序会影响最终判断 p?

十. (5 分) Alice 和 Bob 玩扑克牌比赛，比赛包含多局. 每局 Alice 赢的概率都是 p，并且各局之间是相互独立的. 他们事先规定了，谁先多赢两局，谁就赢得整个比赛，比赛也就此结束. 请求出 Alice 赢得整个比赛的概率. 请写出你的求解过程.

清华大学本科生考试试题专用纸

Course Title: Probability Theory Time and Date(UTC+8): 13:30-16:05 Nov. 11, 2020

Name:_____ Student ID:_____

Note: This exam is closed book and closed notes. There are 10 problems, worth 100 points in total.

1. (18 points) True or False Questions: In the probability space $(\Omega, \mathscr{F}, \mathbb{P})$,

(1) (2 points) Given that A is an event, if $P(w) = 0$ for any $w \in A$, then $P(A) = 0$;

(2) (2 points) Suppose random variable X follows a Poisson distribution ($X \sim \mathcal{P}(\lambda)$), then $\mathbb{P}(X = k+1 | X > k) = \mathbb{P}(X = 1)$.

(3) (2 points) If the random vector (U, V) are jointly continuous, then $P(U = V) = 0$.

(4) (2 points) Suppose the random variable $X \sim \mathcal{N}(0, 1)$, and $Y = X$. Then $P(X = Y) = 0$.

(5) (2 points) Given that A,B and C are events, if $P(A \cap B \cap C) = P(A)P(B)P(C)$, then A, B and C are mutually independent.

(6) (2 points) Given that A and B are events, if $A \cap B = \emptyset$, then A and B are independent;

(7) (2 points) If X, Y, Z are mutually independent discrete random variables, then $X + Y$ and Z are independent;

(8) (2 points) Given that A and B are events, if $P(B) > 0$, then $P(A|B) \geq P(A)$.

(9) (2 points) Given that the event A happened, then $P(A) = 1$.

2. (8 points)

Suppose the CDF of a random variable X is as follows:

$$F(x) = \begin{cases} 0, & \text{if} \quad x < -1, \\ \frac{1}{2}x + \frac{1}{2}, & \text{if} \quad -1 \leq x < 0, \\ \frac{3}{4} + \frac{1}{4}x, & \text{if} \quad 0 \leq x < 1, \\ 1, & \text{if} \quad x \geq 1. \end{cases}$$

Please(1)(2 points)draw the CDF, and find: (2)(2 points) $P(X = -1)$, (3)(2 points) $P(X = 0)$, (4)(2 points) $P(X > \frac{1}{2})$.

3. (6 points) Alice plan to go to a bank for some service for 4 times in total. For each time, suppose the time X(unit: minute) she has to wait for the service follows an exponential distribution $X \sim \mathcal{E}(\frac{1}{5})$. The waiting time is mutually independent. Find the probability that Alice waits less than 10 mins at least once. (No calculator required. Just provide the final simplification of the formula.)

4. (10 points)

(1)(5 points) Suppose $X \sim \mathcal{E}(\lambda)$. Find the PDF of $Y = 2X + 1$.

(2)(5 points) Suppose $X \sim \mathcal{U}(-1, 1)$. Find the PDF of $Y = |3X|$.

5. (15 points) Suppose the joint PDF of (X, Y) is as follows:

$$f(x,y) = \begin{cases} 6x^2 e^{-2y}, & \text{if} \quad 0 \le x \le 1, y > 0, \\ \\ 0, & \text{otherwise.} \end{cases}$$

(1)(5 points) Are X and Y independent? Please give your reason.

(2)(5 points) Find the marginal CDF $F_X(x)$.

(3)(5 points) Find the distribution of Y (including the name and parameters).

6. (14 points) A class has N students in total, including N_1 freshmen, N_2 sophomores,N_3 juniors and N_4 seniors separately. $N_i > 0, i = 1, ..., 4$, and $\sum_{i=1}^{4} N_i = N$. Teacher Alice call the roll randomly each time, with all N students equal likely. Suppose Alice call the roll for n times. Denote X_1, X_2, X_3, X_4 as the number of freshmen, sophomores, juniors and seniors in the n times, separately.

(1)(4 points) Find the joint PMF of X_1,...,X_4, i.e. $P_{X_1,...,X_4}(k_1, ..., k_4)$.

(2)(4 points) Find the PMF of (X_1, X_2).

(3)(3 points) Are X_1 and X_2 independent? Please give your reason.

(4)(3 points) Teacher Bob call the roll randomly, but without students that have already been called. For example, suppose Bob picked Kevin first, then Bob will pick from the left $N - 1$ students equal likely, and will not pick Kevin anymore. Denote Y as the number of sophomores that Bob picked, when Bob picked r students in total. $r \le N$. Find the PMF of Y.

7. (9 points) A point is chosen at random (according to a uniform PDF) within a semicircle of the form $\{(x, y)|x^2 + y^2 \le r, y \ge 0\}$, for some given $r > 0$.

(1)(3 points) Find the joint PDF of the coordinates (X, Y) of the chosen point.

(2)(3 points) Find the conditional PDF $f_{X|Y}(x|y)$ of the coordinates X, given $Y = y$.

(3)(3 points) Are X and Y independent? Please give your reason.

8. (10 points) Suppose for some given trial, the sample space $\Omega = [0, 1]$. Denote events $A_n = (\frac{1}{3^n}, \frac{1}{3^{n-1}}]$, $n = 1, 2, ...$

(1)(4 points) Find the σ-field $\mathscr{F}$ that is generated by $\{A_2\}$.

(2)(3 points) For each event A, define the probability $\mathbb{P}(A)$ as the interval length. Find $P(\lim\limits_{n\to\infty} \cup_{i=1}^{n} A_i)$.

(3)(3 points) Find $\limsup\limits_{n\to\infty} A_n$.

9. (5 points) For some given disease A, assume its prevalence rate in the population is 0.1. Now two hospitals provide tests with "90% accurate" separately: if a patient

has the disease A, then the test result shows positive with 0.9 probability; if a patient doesn't have the disease A, then the test result shows negative with 0.9 probability. Also assume that given whether a patient has the disease, the two test results from the two hospitals are conditionally independent.

(1)(3 points) Find the probability that the patient has the disease A, given that both hospitals provide positive test results.

(2)(2 points) Bob went to the west hospital first and got negative result. Then he went to the east hospital and got positive result. Considering the probability that he gets the disease given these facts, he regrets that he should go to the east hospital first. Please help him decide whether the order influence the probability that he gets the disease.

10. (5 points) Alice and Bob play a card match. This match consists of a series of independent games. Alice has probability p of winning each game. Before the match, they made a rule: the first player to win two games more than his opponent wins the match. Please find the probability that Alice wins the match.